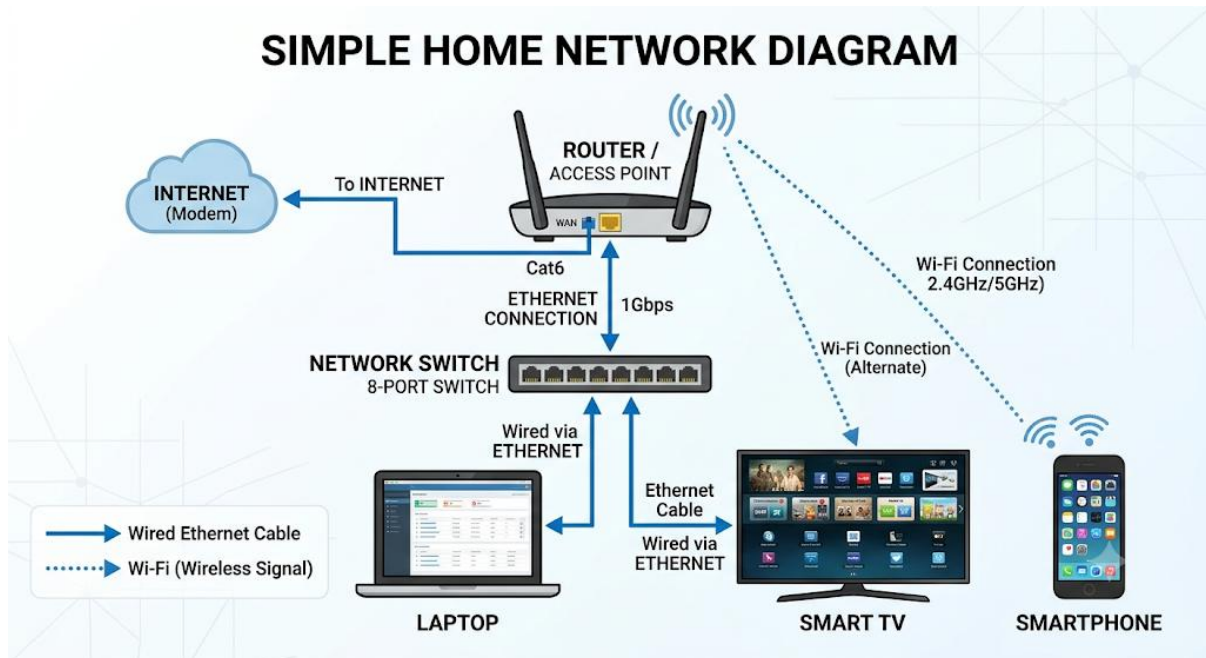


Network Devices

Task (1):



Task (2):

Network Device	Main Function	Real-World Use Case	Daily App/Service Example
Router	Connects different networks and directs data between them.	Connects your home devices to the Internet.	YouTube: Your router sends your phone/laptop's video requests to the Internet for streaming.
Switch	Connects multiple devices within the same LAN and forwards data to the correct device.	Connects computers, printers, and servers in a college computer lab.	LAN Gaming: Multiple PCs can connect through a switch to play games together on the same local network.
Hub	Connects multiple devices in a LAN and broadcasts data to all connected devices.	Used in simple or older small networks for connecting computers.	File Sharing: Connected PCs can communicate over a basic local network, although hubs are less efficient than switches.
Access Point (AP)	Provides wireless Wi-Fi connectivity to devices and connects them to a wired network.	Provides Wi-Fi coverage in homes, offices, colleges, and public places.	Wi-Fi Streaming: Your smartphone or Smart TV connects through an AP to stream YouTube videos.

Task (3):

❖ Scenario: Ordering Food on Zomato:

- Imagine you use your phone to order food through **Zomato**. First, your phone connects to the home **Wi-Fi Access Point**, which provides the wireless connection.
- The Access Point sends your data to the **router**, which forwards the request from your home network to the Internet.
- As the request travels through the Internet, **switches** forward the data between different network devices and servers until it reaches Zomato's servers.
- Zomato processes your order and sends the confirmation back through the same network path to your phone.

Data flow:

Phone → Access Point → Router → Internet Switches → Zomato Server → Back to Phone

Task (4):

❖ Two Ways to Connect the Gaming PC to the Wi-Fi Router:

Network Device	How It Works	Setup
1. Wi-Fi-to-Ethernet Bridge	Receives the router's Wi-Fi signal and converts it into a wired Ethernet connection for the PC.	Router  → Wi-Fi Bridge → Ethernet Cable → Gaming PC 
2. Powerline Adapter	Uses your home's electrical wiring to carry the network signal from the router to another room, where the second adapter provides Ethernet.	Router → Powerline Adapter → Electrical Wiring → Powerline Adapter → Ethernet → PC

Recommendation: A **Wi-Fi-to-Ethernet bridge** is usually simpler to set up. For gaming, a **powerline adapter** can be useful when the PC is far from the router and Wi-Fi signal is weak.